Count Me In TT Teacher Guide

CountMeInTT Comprehensive Teacher Implementation Guide

Prepared for Educators in Trinidad & Tobago

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1. Executive Overview

Count Me In TT (CountMeInTT) is a free, web-based math fluency platform that helps primary students build speed and accuracy with multiplication facts through timed grid challenges . It transforms routine multiplication practice into a game-like challenge: students race to fill in a multiplication table grid as fast as they can, competing primarily against their **own personal best times**. The platform logs each attempt and provides instant feedback on accuracy, requiring students to correct any errors before finishing. When a student completes a grid and submits their result, their time can (optionally) appear on national, school, and class leaderboards, adding a layer of friendly competition and recognition .

For teachers, CountMeInTT offers a plug-and-play classroom routine that produces **measurable progress evidence** with minimal marking or preparation . A typical implementation involves either a brief daily practice or a weekly fluency session (around 10 minutes per class, on average). The platform automatically checks answers and records times, so teachers are freed from grading dozens of drills; instead, they can focus on **facilitating** the activity and encouraging students. Over time, each student's improvement in speed and accuracy is clearly documented through their recorded times, providing data that can be used for formative assessment and to celebrate growth.

In summary, CountMeInTT addresses a key foundational skill – multiplication fact fluency – in an engaging, efficient way. It is designed to slot into the existing school timetable with minimal disruption, while tackling common classroom challenges such as uneven multiplication mastery, math anxiety under time pressure, limited device availability, and tight instructional time . By regularly using CountMeInTT, students gradually achieve automaticity in their times tables, which in turn frees up their working memory for more complex problem-solving. Teachers benefit from a consistent routine that not only improves basic math skills but also builds student confidence and a positive attitude toward mathematics.

2. Introduction: The Math Fluency Challenge in Trinidad & Tobago

Mathematics educators in Trinidad and Tobago are acutely aware of a persistent problem: many primary school students struggle with **basic multiplication fluency**, hindering their overall math performance. By Standard 4–5, pupils are expected to have mastered their times tables, yet in practice there is often **uneven times-table mastery across a class** . It's not uncommon to find students in Standard 5 who still count on fingers or use lengthy methods to multiply, which slows them down significantly in exams and more advanced calculations. This lack of quick recall can contribute to errors, frustration, and a loss of confidence in math. When these students face the timed conditions of exams like the Secondary Entrance Assessment (SEA), their anxiety is compounded – they not only have to solve problems but also recall facts under pressure.

The impact is evident in national assessment outcomes. In recent years, average performance in the SEA Mathematics exam has hovered around only 50% . For example, in 2024 the average math score was about 50.7%, essentially unchanged from the previous year and a modest improvement from the even lower averages during the pandemic period . While many factors affect these results, foundational skills such as multiplication fact fluency play a role. Students who have not internalized their multiplication tables often struggle with higher-level topics like fractions, division, and problem-solving, which require fluent recall of basic facts. The Ministry of Education has acknowledged “challenges in some strands of mathematics” and emphasized targeted support in the upper primary years to address learning gaps . Ensuring that students enter secondary school with core competencies – including basic fact fluency – is crucial, as pupils without this competency “will find some difficulties” in secondary-level math .

Several classroom realities contribute to the fluency gap. **Time and curriculum pressure** make it difficult for teachers to allocate enough practice time for every student. Under the tight syllabus, teachers may give homework worksheets or occasional timed quizzes on multiplication, but **marking these by hand is time-consuming**, leading to infrequent feedback for students . In class, when a few students answer quickly and others lag behind, the slower students often become anxious or disengaged. **Math anxiety** can appear early – some children start fearing math when they feel they are “slow” or always the last to finish . Research has shown that such anxiety can start as early as primary level and can negatively affect working memory and performance . In Trinidad & Tobago, as elsewhere, timed arithmetic tests have sometimes been viewed as “torturous” exercises that create stress rather than fluency if done improperly . This anxiety undermines learning: when a child is anxious about remembering 7×8, the stress literally takes up mental bandwidth (working memory) that could otherwise be used to solve the problem . Thus, the very act of recalling facts under pressure becomes a hurdle.

Another challenge in the local context is **limited resources**. Not all schools or homes have access to a plethora of devices or internet at all times. Teachers may have to manage with a single classroom computer or a few tablets for activities beyond the textbook. This reality can constrain the use of fancy math software or one-to-one digital interventions. What’s needed are **flexible solutions that can work with just one device or many**, fitting the diverse scenarios in our primary schools. Moreover, any extra activity introduced by a teacher must be **low-workload and sustainable** – if it feels like “one more thing” adding to teachers’ already heavy load of lesson planning, grading, and co-curricular duties, it won’t be adopted consistently .

In summary, Trinidad & Tobago’s primary math classrooms face a convergence of issues: **gaps in basic multiplication fluency, math anxiety under time pressure, limited practice time, and resource constraints**. The combination of these factors means that traditional methods (like rote memorization or sporadic timed tests on paper) often fall short of solving the fluency problem. There is a clear need for a **practical, scalable approach** to help all students gain automaticity in multiplication facts in a way that is engaging and non-threatening to their confidence. This introduction frames the backdrop against which CountMeInTT was developed – as a response to the question: *How can we help every student get faster and more confident in math, without adding to teachers’ burden or students’ anxiety?*

3. Research Insights on Math Attitudes and Performance

Efforts to improve math outcomes benefit from understanding the research on how students feel about math and how they learn best. In Trinidad and Tobago, studies have highlighted significant **affective factors** influencing math teaching and learning. A 2017 national survey of mathematics in primary schools (conducted by NIHERST) found that teachers frequently encounter difficulties such as varying student readiness, lack of time for individualized practice, and the challenge of keeping students engaged in repetitive drills . The same survey underscored that **practical constraints in the classroom** – like large class sizes and limited class time – often push teachers to focus on completing the syllabus rather than reinforcing fundamental skills through repeated practice. Teachers expressed a need for more efficient methods to develop skills like times tables without sacrificing precious instructional time on manual grading or one-on-one drills . These findings support the idea that a quick, repeatable tool which requires **minimal marking and preparation** can be invaluable in real classrooms.

Another local study, conducted at the University of Trinidad and Tobago (UTT) in 2013, examined the attitudes of primary **student-teachers** (i.e. trainee teachers) toward mathematics . The results were telling: many trainee teachers themselves harbored anxieties and lack of confidence in math, often stemming from their own school experiences. These affective factors inevitably influence how they will teach math. If a teacher is uneasy with math or believes that only some students can excel, those attitudes can subtly transfer to the classroom environment. The UTT research highlights the importance of approaches that build **confidence and positive attitudes** – not just in students, but in teachers as well. This is directly relevant to CountMeInTT's emphasis on enjoyment, confidence, and persistence; a tool that makes math practice feel like a fun challenge can help reshape negative attitudes. When teachers see their students *enjoying* a multiplication activity and cheering each other on, it can also shift the teacher's mindset about math instruction from punitive speed drills to positive skill-building.

Beyond our borders, regional and international research align with these observations. The Organisation of Eastern Caribbean States (OECS) curriculum standards, which are closely related to the Caribbean Primary Exit Assessment expectations, emphasize developing **fluency foundations** in the primary years . Fluency (in addition, subtraction, multiplication, and division facts) is seen as a critical step in the progression toward higher-order competence. In other words, education experts in the region agree that *knowing your tables* is a gateway to success in more complex math. The OECS standards underscore regular practice and mastery of basic facts by around Grade 4, reinforcing that our focus on multiplication fluency in Std 2–5 is well-placed.

Cognitive science research has further illuminated *why* fluency matters. When basic facts are automatic, students free up mental resources for problem-solving. Studies by Ashcraft and colleagues demonstrated that **math anxiety can compromise working memory**, which is vital for multi-step calculations and reasoning . A student who is nervous and still figuring out 7×8 in the middle of a long division problem is expending brain power on that simple retrieval, leaving less capacity for the larger task. Over time, consistently struggling with such basics can lead to

entrenched math anxiety and avoidance of math tasks . Conversely, building strong fact fluency can alleviate anxiety by instilling confidence – students know they can rely on their instant recall and are not as fearful of being put on the spot.

It's worth noting that the **format** of practice also matters for attitudes. Traditional timed tests have a bad reputation, sometimes deservedly. When used solely to rank students or delivered without support, they can exacerbate fear. As Jo Boaler (2012) noted, research has linked relentless, high-pressure timed testing with the early onset of math anxiety in children . However, emerging evidence suggests that *not all timed practice is inherently harmful* – it depends on how it's implemented . When timed challenges are framed as personal games or low-stakes activities (rather than high-stakes tests) and when students are supported to reflect on their progress instead of competing negatively, the experience can actually be empowering. The key is to avoid making speed the **sole measure of math ability**. CountMeInTT's design is informed by these insights: it shifts the timed practice paradigm from a stressful test to a fun challenge, emphasizing personal improvement and effort over comparison.

Finally, extensive research in educational psychology supports the use of **regular, spaced practice and immediate feedback** to build mastery. These principles will be discussed in the next section, but it is important to note here that the approach CountMeInTT takes is grounded in evidence-based strategies for effective learning. In sum, both local findings and global research point to a solution that is *regular, confidence-building practice* of math fundamentals in a positive environment. CountMeInTT was developed with exactly that in mind – to tackle the fluency problem by blending solid learning science with practical classroom know-how.

4. Overview of the CountMeInTT Program

4.1 Origin and Goals of CountMeInTT

CountMeInTT was born out of on-the-ground experiences in Trinidad & Tobago's education system. Its creator, Andre Burton (a local math tutor and educator), noticed a recurring pattern: **students who basically understood math procedures still struggled with speed, confidence, and recall**, especially under exam conditions . Many pupils would get flustered by simple multiplication or take far too long on routine calculations, leaving them less time and mental energy for complex problems. Over years of tutoring, he observed that this often led to a fear of numbers that could persist into secondary school and beyond . Determined to address this "serious reality," Burton envisioned a tool that would make fundamental practice engaging and motivating, instead of tedious or intimidating.

The goal of CountMeInTT is to **make multiplication practice feel like play rather than work** . As Burton explains, he wanted to build something that students would *actually want* to use repeatedly – capturing some of the excitement of a game or challenge. The philosophy is simple: when practice is enjoyable and students can visibly track their own improvement, they will voluntarily put in the repetitions needed to build fluency. Every feature of CountMeInTT was designed with a dual focus on **academic development** and **positive emotional experience**.

The platform aims to strengthen core math skills (speed and accuracy in multiplication) *and* to develop traits like focus, perseverance, and confidence through gameplay .

Importantly, CountMeInTT was also conceived as a **free and accessible resource** for all students, teachers, and schools. It's provided at no cost so that no child is left out due to financial or technological barriers . The only requirements are a device with a web browser and an internet connection – which many schools have at least in some capacity. By keeping it web-based and lightweight, the program avoids the need for special software or expensive hardware. The inclusive goal is clear: *"Every child deserves access to tools that can strengthen their foundation in mathematics, regardless of their background or resources"* . In summary, the origin and goals of CountMeInTT are rooted in local educational needs – it's a tool built in Trinidad & Tobago, for Trinidad & Tobago – aiming to uplift math fluency and attitudes through an enjoyable, challenge-based approach.

4.2 How the Game Works (Design and Mechanics)

CountMeInTT in a nutshell: it is a timed multiplication grid game. When a student opens the CountMeInTT website (on a computer, tablet, or smartphone), they are greeted with a simple interface to **choose a grid size** and start the challenge . The available grid levels are:

- **5×5 Grid (Quick)** – a 5-by-5 multiplication table covering facts up to 5×5.
- **5×12 Grid (Trainer)** – a 5-by-12 grid covering facts up to 5×12 (i.e., one multiplicand 1–5 and the other 1–12).
- **12×12 Grid (Classic)** – the full traditional times table up to 12×12.
- **15×15 Grid (Pro)** – an extended challenge grid up to 15×15 for high achievers or secondary students.

Students typically use whatever grid size the teacher or their level recommends (see Section 5.1 for guidance by class level). Once a grid is selected, the **game screen** presents a multiplication table layout with empty cells. It's essentially a blank multiplication chart that the student must fill in with the correct products for each multiplication fact. For example, if using the 5×12 grid, there will be 5 rows and 12 columns of facts (often these include 1× through 5× for rows, and ×1 through ×12 for columns, covering tables 1–12 for the numbers 1 to 5). The exact layout (whether factors are labeled or randomized) isn't complex – the key point is the student has a finite set of multiplication questions to answer as quickly as possible.

Timing and input: The moment the student starts typing the first answer, the **timer starts automatically** . The timer is prominently displayed, creating a sense of challenge to finish the grid faster. Students then proceed to fill in each cell with the correct product (e.g., if one cell represents 7 × 8, the student would enter 56). They can move cell by cell, typically using the tab or arrow keys or tapping on the next cell on a tablet. **If an answer is correct, the cell turns**

green; if incorrect, it turns red . This immediate color feedback is important – it alerts the student to mistakes right away. Notably, the game *requires* that all answers be correct for the grid to be considered complete; **students must correct any errors to finish the game** . This design ensures that the focus is on both speed *and* accuracy. A student cannot just blitz through typing random numbers; they must pay attention and fix wrong answers, which reinforces learning of the correct fact.

Some friendly touches in the interface include highlighting **perfect squares in a special color (gold)** . For instance, 6×6 or 9×9 might appear in gold font or with a star – this subtly teaches students about square numbers as they notice those cells. It's a minor gamified element that can spark curiosity ("Why are those special?").

Finishing and submitting: The timer continues until every cell is filled correctly. **The timer stops only when all answers are correct** , marking the completion of the attempt. At this point, the student is prompted to **submit their result**. The submission involves entering a name (or nickname), selecting their school from a list or typing it, and their class level (Std 2, Std 3, etc.) . There is also an optional email field, primarily used if there are prizes or to prevent duplicate names, but it's not required for classroom use . Students **must enter a name and school/class** in order for their attempt to count on the leaderboards. Once they hit submit, the data (name, school, class, grid level, and time) is logged in the CountMeInTT database and the student can immediately see how their time compares on the **leaderboard**.

The **leaderboards** are accessible on the website and show top times in various categories:

- A **Global Leaderboard** displays the top performers across Trinidad & Tobago by grid type (with filters for category or age if applicable) . It shows a "podium" of the three best times and a ranking of other fast times, along with the student's name (or nickname), school, and time .

- **School-specific and Class-specific Leaderboards**: The platform also features individual pages for each school, listing all submissions from that school and sorting by best times . Within that, it can list by class level. In practice, this means a teacher can ask students to enter the school name *consistently* and their class (e.g., "GhazPersad Primary – Class 4A"), and then view a page that shows just the results for 4A of GhazPersad Primary. This is extremely useful for **motivation and monitoring** at the class level – students love seeing their name move up the class leaderboard as they improve, and teachers can quickly spot who is making progress or who might need help.

Behind the scenes, CountMeInTT also logs every attempt (not just the top times). For example, if a student plays ten times, all ten results are stored (though only their best might show on the public leaderboard). This data can be used to produce **progress reports** for a teacher or principal. Indeed, the platform offers optional reporting: for instance, a teacher or school can request a weekly summary of all their students' times, or a **snapshot** of class averages and

improvements each month . These reports are typically provided via email or downloadable files and can highlight metrics like number of attempts, best improvements, and so on.

In terms of **accessibility**, CountMeInTT was built to be device-agnostic:

- On **smartphones**, it's recommended to rotate to landscape orientation for a larger keyboard, and use the numeric keypad for faster entry .

- On **tablets**, the screen size is usually sufficient and allows fairly quick typing – tablets are often an ideal device in classrooms for this game .

- On **laptops/desktops**, students can use the keyboard which tends to yield the fastest typing speed; these devices are great for competitive scenarios or whole-class play .

  In any case, the interface is simple and has virtually no learning curve – even Standard 1 or 2 students can grasp what to do after a brief demo.

What CountMeInTT is not: It is not a comprehensive math teaching program or a curriculum replacement. It focuses narrowly on multiplication fact recall (and implicitly, on some related skills like pattern recognition). It doesn't teach *concepts* from scratch – rather, it assumes that students have been introduced to multiplication and need practice to gain fluency. It's best used as a **supplementary tool**, a kind of modern "times table drill" with enhancements. Recognizing this, the design avoids distractions: there are no elaborate storylines or unrelated mini-games; the reward comes from beating one's own time and climbing the leaderboard in a meaningful skill. This simplicity is deliberate to keep the practice efficient and aligned with its core purpose.

In summary, the mechanics of CountMeInTT are straightforward: **select a grid, race the clock to fill it accurately, submit your time, and see your ranking**. This simplicity belies the thoughtful pedagogy underneath, which we explore next, but it's key to the tool's classroom success that it's easy to run and easy for kids to understand. A teacher can explain the rules in a minute, even project a sample and do a quick demo, and then let students try – the game itself guides them with the timer and color feedback.

4.3 Learning Principles and Pedagogical Value

CountMeInTT may appear to students as just a fun game, but it was crafted around proven learning principles to ensure that it delivers genuine educational value. Here we highlight the key pedagogical benefits and the research basis for each:

- **Retrieval Practice (The Testing Effect):** One of the strongest findings in cognitive science is that actively retrieving information (as opposed to just reviewing it) strengthens memory and learning. Each CountMeInTT attempt is essentially a form of retrieval practice – students must recall multiplication facts from memory, repeatedly. According to Roediger & Karpicke (2006), taking memory tests (i.e. practicing retrieval)

significantly improves long-term retention of information . In our context, when a student pulls up “7×8 = 56” from memory under a bit of time pressure, it reinforces that fact in their brain’s long-term storage. Over weeks of doing this, the multiplication facts become more and more automatic. CountMeInTT leverages this effect by encouraging **frequent recall in a low-stakes environment**. It’s like giving a quiz that students actually enjoy, thereby harnessing the testing effect without the stress of grades.

- **Spaced & Distributed Practice:** Research by Cepeda et al. (2006) and others shows that spreading out practice over time (as opposed to cramming it in one session) leads to better retention . CountMeInTT is built for distributed practice – a few minutes each day or each week, throughout the term. Instead of doing a massive drilling session once a term, students using the program get regular, short practice. This spacing means each recall session strengthens memory and the forgetting curve is continuously reset. The guide recommends 5–10 minutes of weekly use, which naturally implements spaced repetition. By having, say, one attempt per student each week, across a term of 10 weeks, a student effectively practices those facts 10 times on separate occasions – a recipe for durable learning.

- **Immediate Feedback & Error Correction:** The game’s design gives **instant feedback** on each answer (green for correct, red for incorrect), and crucially, it forces the student to correct errors to finish. This is a form of formative feedback loop: students learn from mistakes in the moment. Immediate feedback is known to help in building skills, as it prevents reinforcement of wrong information and guides students toward the correct answer while their thought process is still fresh. In contrast to paper drills where a student might practice 20 questions and only find out later which ones were wrong (or never get corrections if not reviewed), CountMeInTT’s approach means a student cannot complete the task without knowing and fixing the right answer. This turns every mistake into a teaching moment *on the spot*. Educational psychology identifies this kind of feedback as highly effective for procedural learning – it’s akin to having a personal tutor that says “try again, not quite right” immediately as the student works.

- **Fluency and Automaticity:** Why focus on speed? It’s not about rote memorization for its own sake, but about achieving **automaticity**, where basic operations don’t consume mental effort. When multiplication facts are automatic, a student’s cognitive load in complex tasks is reduced. They can tackle multi-step problems or word problems without being bogged down by basic calculations. As cited earlier, research by Ashcraft & Krause (2007) indicates that freeing up working memory by automating lower-level skills can improve overall math performance and reduce anxiety . CountMeInTT specifically targets this: by making students faster at times tables, it effectively frees up “brain space” for reasoning and problem-solving. Teachers often find that once students stop struggling with times tables, their performance in fractions, division, algebra and so on improves because they aren’t hindered by slow fact recall.

- **Goal-Setting and Self-Monitoring:** A subtle but important pedagogical aspect is that CountMeInTT encourages students to **beat their own best time** rather than simply aiming to be faster than others. This fosters a form of goal-setting and self-monitoring. Students see their personal record and naturally aim to improve it next time. Teachers can enhance this by having students write down their times and set a target (e.g., "I got 3:00 this week, I'll try for 2:30 next week"). This practice ties into research on self-regulated learning – when learners set goals and track progress, they tend to be more motivated and achieve more. Indeed, the design aligns with known motivational techniques: it's **gamification with a purpose**. The use of leaderboards, personal bests, and "levels" (grid sizes) taps into gamification elements that have been shown in meta-analyses to produce modest but positive gains in student learning outcomes . The key is that the competition is positive and **progress-oriented**. Students are not labeled as "bad at math" if they are slower; instead, even the slowest student can feel success by improving their own time by, say, 20%.

- **Psychological Safety and Growth Mindset:** CountMeInTT's structure supports a growth mindset by emphasizing improvement and effort. The platform itself does not stigmatize slower times – a student's time is not used for a grade or punishment, and leaderboards can be viewed in categories (so a Standard 3 student isn't directly competing with a national top Standard 5 student on the same board – they look at class or school rankings). Teachers can further ensure the environment is safe by how they frame the activity (more on that in Section 6.3). Because the game is repeatable, students learn that **practice leads to progress**: it is very tangible when a child goes from taking 5 minutes to complete the grid down to 3 minutes a few weeks later. This visible growth can be incredibly motivating and helps instill the idea that math ability is not fixed – you can improve with effort. That insight is backed by educational research on mindset and motivation.

- **Minimal Intrusion on Curriculum:** While not a learning principle per se, it's pedagogically significant that CountMeInTT is *not* trying to teach new content but reinforce what has been taught. This aligns with the idea of **spiral review** or maintenance of skills. A teacher using CountMeInTT doesn't have to restructure their curriculum – they can plug this in as a warm-up or a filler whenever there's a spare 10 minutes. It reinforces what the curriculum already covers (times tables by Std 3/4) and does so in a manner that integrates with class routines (for example, as part of mental math practice sessions).

In essence, the pedagogical value of CountMeInTT comes from marrying these powerful learning strategies with an implementation that is extremely practical for the classroom. The program is more than a game – it is a **fluency-building system** that employs retrieval practice, spaced repetition, immediate feedback, and motivational gamification to help students achieve automaticity in multiplication. It aligns with best practices (as summarized in Dunlosky et al.,

2013, which recommends practice testing and distributed practice as top-tier study techniques) in a package that students actually like.

Furthermore, by logging progress and making improvement visible, it provides teachers with *data* and students with *reinforcement*. A principal or curriculum coordinator can appreciate that using this tool regularly generates evidence of learning – something that can be quite useful for evaluations and accountability, without imposing testing stress on kids or marking stress on teachers. Finally, the pedagogical underpinning ensures that CountMeInTT is not just a gimmick; it's a **strategic intervention** addressing a critical bottleneck in math education with methods that research shows "work." The next sections will delve into how to implement this tool in the real-world classroom context to maximize these benefits.

5. Classroom Implementation Strategies

This section provides a practical guide to implementing CountMeInTT in a classroom setting. It covers what grid level to use for each class, how to structure the weekly routine, how to manage with various numbers of devices, and tips for everyday classroom use. The strategies are designed to be **flexible** – whether you have one device or one for every student, whether you prefer daily practice or a weekly session, you'll find a model that can work for you. The overarching principle is to ensure **each student gets at least one meaningful attempt per week** on the platform, in a way that's manageable for the teacher and engaging for the students.

5.1 Recommended Grid Levels by Class (Std 2–5)

CountMeInTT offers multiple grid sizes (5×5, 5×12, 12×12, 15×15). Choosing the right level for your students is important to hit the sweet spot of being **challenging yet attainable**. Here are the recommended grid levels for each primary class, based on both the typical curriculum stage and evidence gathered from pilot implementations:

- **Standard 2:** Start with the **5×5 grid** (Quick). This smaller grid is an excellent confidence-booster and entry point for young learners . At Standard 2, many students are still mastering the 2×, 3×, 4×, 5× tables. The 5×5 grid focuses on lower times tables and keeps the number of questions manageable (only 25 facts). Using the 5×5 grid weekly allows Std 2 students to build foundational recall without being overwhelmed. It's common that by the end of Std 2, students might "graduate" to a 5×12 grid if they've excelled, but initially 5×5 gives everyone a fair shot at completing the grid in a reasonable time (a few minutes).

- **Standard 3:** Begin the year with **5×5 grids**, and as the class improves, progress to **5×12 (Trainer)** for most students . Standard 3 is a transitional year where the goal is full mastery of 1–12 times tables. Early in Std 3, many students may still be shaky on the higher tables (6×, 7×, 8×, 9×). Starting with 5×5 ensures they get faster on the lower half first. As you observe that the majority of the class can comfortably finish a 5×5 grid in, say, under 2 minutes or so, you can introduce the 5×12 grid. One effective strategy is

team-based play when first introducing the larger grid: e.g., split the class into teams that collaboratively fill a 5×12 grid (with each student responsible for certain rows) – this way they learn patterns from peers (see Section 6.3 for using team play to support hesitant students) . By the end of Std 3, ideally students are using the full 5×12 grid in individual play, setting them up for success in Std 4.

- **Standard 4:** Use the **5×12 grid** as the default weekly challenge . At this level, all students should be working with the complete 1–12 multiplication range regularly, as SEA preparation is on the horizon. The 5×12 grid (60 questions) practiced weekly will solidify their recall. In addition, it is recommended to **introduce the 12×12 grid occasionally as a structured challenge** . For example, once a month or once a term, you might hold a "Classic Grid Challenge" where students attempt the full 12×12 grid (144 questions). This is a stretch goal to push fluent students further and to give everyone exposure to a more intense challenge under supportive conditions. However, do this without pressure – perhaps in a fun contest format or as a special event – since not all Std 4 students will be ready to complete 12×12 quickly. The mainstay remains 5×12 for fluency, with 12×12 as an enrichment activity or for the quicker finishers as extra credit.

- **Standard 5:** Continue with **5×12 grids on a weekly basis**, as these cover the core facts needed for the SEA exam mental math and general arithmetic . By Std 5, many students will achieve very fast times on 5×12 with practice (some even under 2 minutes). At this stage, incorporate the **12×12 grid regularly**, perhaps every other week or in alternating cycles . If devices/time allow, you might run both a 5×12 and a 12×12 attempt each week (the former as a warm-up and the latter as the official one, or vice versa). The 12×12 grid ensures they truly have full multiplication mastery and builds endurance. Additionally, the **15×15 grid (Pro)** can be used as an extension for advanced students or as a "fun challenge" for the whole class after SEA . Some Std 5 teachers use the 15×15 grid post-exams when students are essentially done with the curriculum and need enrichment – it's a way to keep them engaged with math in a playful manner. However, 15×15 is not expected from everyone; it's more of a bonus level to prevent boredom for those who have excelled.

To summarize the recommendations:

- *Std 2:* **5×5** (focus on building confidence and core patterns)

- *Std 3:* **5×5**, moving to **5×12** when ready (full tables by year's end)

- *Std 4:* **5×12** weekly; introduce **12×12** occasionally for challenge

- *Std 5:* **5×12** weekly; **12×12** frequently (alternate or weekly); **15×15** for high-flyers or bonus practice

These are guidelines, not rigid rules. A teacher should gauge the class's comfort. If a Standard 4 class is struggling, there is no harm in sticking to 5×12 all year and only doing a couple of 12×12 attempts towards the end. Conversely, if a Standard 3 class is very strong, you might accelerate them to 12×12 earlier (perhaps in team mode). The overarching goal is to use a grid level that ensures **success with effort** – students should generally be able to finish (even if slowly) and then work on improving that completion time week by week. By aligning the grid difficulty with class level, you ensure the activity remains in that sweet spot of *productive challenge*.

5.2 Routine Models: Daily Micro-Practice vs Weekly Fluency Block

Different schools and teachers have different scheduling realities. CountMeInTT can be implemented flexibly, but two broad routine models have emerged as the most practical and effective:

A) 10-Minute Daily Micro-Routine (Mon–Fri) – This model breaks the class into small subsets and spreads the activity across the week. It is **ideal when devices are very limited** (e.g., only 1 or 2 computers in the class) or when you want to make fluency practice a daily habit in bite-sized chunks . Here's how it works:

- Allocate about 10 minutes each day for CountMeInTT as a warm-up or conclusion to the math lesson.

- Each day, a **small group of students (or pairs)** gets to do their official attempt, while others either observe or do a quick alternate task.

- For example, in a class of 25 with one device: you might have 5 students (in pairs or individually) use the device on Monday, another 5 on Tuesday, and so on, so that by Friday all 25 have had a turn. In this rotation, each student plays once per week, which meets the goal of one attempt per week.

- While 5 students are playing (perhaps 2 at a time if you can manage pairing on one device), the rest of the class can be engaged in a mental math question, a quiet reading, or even acting as "coaches"/spotters (more on the Pair system in 5.4). Some teachers use a station approach where one station is the computer and others are worksheets or other games, rotating groups through over the week.

- This micro-routine requires discipline to stick to the 10 minutes and rotate fairly, but it **minimizes idle time** for devices. It's a great solution for classes with only a single laptop or tablet.

The daily routine has the benefit of *frequency*: students get a constant reminder of math practice, and it becomes a daily expectation. It also smooths out the device demand – you never need more than one or two devices at once. Over a week, everyone gets their turn. One

thing to plan is how to select the groups each day (alphabetical order, or maybe have a chart to tick off names). Consistency is key so no one is skipped.

B) 20–30 Minute Weekly Block (Whole-Class or Stations) – This model dedicates one lesson period per week (or bi-weekly) purely to fluency practice. It works best if you have enough devices to get through the whole class in that session, and if your timetable allows carving out a longer block less frequently . Here are two variations:

- **Whole-Class Simultaneous Attempt:** If you have **6+ devices or a full computer lab** available, you can have the entire class attempt the grid at the same time in a 20-minute session . For example, every Friday first period, you bring the class to the lab or bring out the school's set of tablets, and everyone logs on to CountMeInTT and races. The teacher starts the session ("Ready, set, go!") and monitors while all students play. Within a few minutes, the fastest students start finishing; within 10 minutes, most are done (slower ones may take a bit longer, but recall that the platform has no time cap – everyone is allowed to finish at their pace). This scenario is exciting and efficient – it simulates a competition atmosphere, and then you can announce the class "Top 3 and Most Improved" right after. You only need to do minimal coordination (ensuring devices are charged or the lab is booked). A **single weekly block** like this suffices as the official attempt, though students can be encouraged to practice at home as well.

- **Station Rotation in One Block:** If you have a **moderate number of devices (2–5 devices)**, you can still accomplish all attempts within a single larger time block by using **stations or waves** . For example, in a 30-minute session with 5 tablets, divide your 30 students into 5 groups of 6. In the first 6 minutes, group 1 uses the devices (one student per device, or pairs if you prefer coach/player teams), while groups 2–5 are engaged in other mini-activities (flash cards, quick problems, etc.). At the 6-minute mark, rotate: group 2 takes over the devices for the next 6 minutes, and group 1 rotates to a different station or activity. Continue until all groups have played. This approach is akin to running multiple "heats" of a race. It requires some classroom management to keep the other students productively busy (perhaps working on a review worksheet or solving a puzzle) while they wait for their turn, but it can be very effective. Students know that when it's their station's turn, they must focus and perform their attempt, then move aside. In about 5×6=30 minutes, all 30 students had their official attempt.

- **Hybrid approaches:** Some teachers may not have a solid 20–30 min block in the schedule, so they might do half the class one day and half the next (like two semi-blocks). The idea is to still cluster multiple attempts in one session when devices allow, to build a sense of group momentum.

The weekly block model is beneficial in that it creates a **ritual or event** feel ("Friday Fluency Challenge" or "Times Table Tuesday" – whatever you want to call it). Students often look forward to that time. It's also easier for record-keeping since all times come in on the same day; one can immediately update the class leaderboard or tracker after the session. One non-negotiable note,

however, as highlighted in the guide: **if you use full grids (no time caps), to ensure one attempt per student each week, you either need enough devices for a same-day whole-class attempt OR you must use a rolling rotation across days** . In other words, you can't have a scenario where you only have one device and you try to do everyone in one day – that would take too long or cause some students to miss out. So, the teacher must choose either the daily rotation *or* secure enough resources/time for a one-block completion.

Both routine formats – daily micro or weekly block – have been tested and found to work. You might choose the daily micro-routine if you prefer incorporating little drills into each day's class (and it certainly keeps fluency on the radar constantly). Or you might choose the weekly block if you want to minimize disruption to other lessons and create a focused practice time. **Many teachers in Trinidad have leaned towards the weekly 20-minute block** because it fits nicely as a "mental math" period once a week, and they often coordinate device sharing (e.g., borrowing a few tablets from another class on that day). But if your reality is one device, do not be discouraged – the daily 10-minute rotation is perfectly viable, especially with the Pair System described next.

(Teacher Tip: Whichever model you choose, be consistent at first. The power of this tool comes with regular use. If it's daily, try not to skip days; if it's weekly, treat that appointment as sacred as a spelling test or any scheduled activity. Once the routine is established and students expect it, it runs smoothly and you'll see results. Consistency also helps in producing the progress data you want – one data point every week per student is ideal for observing growth.)*

5.3 Device Scenarios and Classroom Setup

One of the strengths of CountMeInTT is its adaptability to different device availability scenarios. Below we outline what **teachers can do in each scenario**, from having a single device to 1:1 coverage, so that the program can run successfully. These recommendations are drawn from field experience and are summarized in the guide for quick reference :

- **Scenario: Only 1 device available (e.g., one classroom computer or teacher's laptop).**

  Recommended approach: Use the **10-minute daily rotation** model across the week . Pair up students (Player + Coach) and have a few pairs take their turns each day (as described in 5.2). With one device, you can realistically have 2 students (1 playing, 1 coaching) at a time. In 10 minutes, perhaps 2 pairs (4 students) can complete their attempts if each attempt takes ~2-3 minutes plus transition. That gives you about 20 students over a 5-day week. If your class is larger, you may need to extend a bit or have some students do it at another time (e.g., break). The key is **rotation and pairing**. Also, consider projecting the device's screen occasionally to involve the whole class: for example, once in a while do a "demonstration round" where one student plays on the laptop connected to a projector and everyone watches/helps – this is good for teaching strategies and keeping others engaged when device count is low.

- **Scenario: 2 devices.**

  Recommended approach: You have a couple of options. You can either follow the **daily micro-rotation** with two devices (meaning you can handle twice as many students per day, perhaps ~8 students a day in pairs on two devices), or you can attempt a **20–30 minute 'two-lane' weekly session** . The "two-lane" approach means in a single session, two students (or pairs) play at the same time on the two devices. For example, you might split the class into two big groups – one group works on some silent task or group work while the other group, two at a time, uses the devices. After 10-15 minutes, swap the groups. This way, all students get a turn in that session via two parallel streams. Essentially, with 2 devices you can cut the needed rotation time in half. It's also possible to combine strategies: maybe do a mini-rotation Monday through Thursday on one device, and Friday have both devices running in parallel to finish off any remaining students.

- **Scenario: 2–5 devices.**

  Recommended approach: This is a moderate number of devices – enough to attempt a **station rotation within a single session** . As described earlier, you could run the entire class in waves during a 20-30 minute block. With, say, 3 devices and a class of 30, you might do 3 students at a time for 5 minutes each, cycling through 10 groups (that might stretch to a bit more than 30 minutes, but you can trim by having some go simultaneously). If you have 5 devices, 5 students can go at once, so probably 6 rounds of 5 students will cover 30 (which fits in around 25-30 minutes if each round is ~5 minutes including switching). **Station organization** is crucial: label devices or stations (Station 1–5), assign students to a station group, use a timer, and have clear signals to rotate. Many teachers with 3-5 tablets have successfully done this by turning it into a game in itself ("Can Station 1 beat Station 2's average time?" etc.). The teacher's role is to supervise both the device stations and the off-device stations, ensuring things move timely.

- **Scenario: 6–10 devices.**

  Recommended approach: At this level of device availability, you can likely have a **whole-class weekly attempt in one go** . For example, if you have 10 tablets in a class of 30, you could run two rounds: 10 students attempt while others watch/cheer quietly (or do an assignment), then the next 10, then the final 10. Each round might take 5 minutes, so in 15 minutes everyone is done. Or if you have closer to 10-15 devices, you might get the whole class (or nearly all) simultaneously, which is ideal. Essentially, 6+ devices bring you close to a one-shot session which is simpler to manage. One thing to note in this scenario: ensure that when multiple students are playing at once, **the teacher actively supervises** to confirm students are entering their names and class correctly in the submission (especially important when many submissions happen concurrently) . You don't want half the class accidentally logging times under the wrong class name or

without a name. Having a routine like “everyone show me your submit screen before you hit submit” can be helpful.

- **Scenario: 1:1 devices (a device per student, e.g., in a BYOD environment or computer lab).**

  Recommended approach: This is the easiest scenario to implement – you can do **whole-class attempts weekly** with ease . Simply schedule a time (e.g., every Wednesday at 10 AM, each student with their tablet/laptop ready, or head to the lab) and let it run. All students start together and you supervise the room, giving encouragement, ensuring no one is copying (not that copying is feasible because of random positions of answers). With one-to-one, you also have the luxury to incorporate **additional practice or challenges**: for instance, you could allow students a second attempt if time permits (maybe one “practice run” and one “official run”), or let early finishers try a harder grid. Moreover, you can encourage students to also use the platform at home since each has their own device – in class you could then highlight those who practiced outside by improvements seen. With 1:1, consider having **headphones available** or ask for silence during play, because the furious typing of 30 students can get noisy and possibly distracting to each other. Generally, though, classes adapt to it like a computer-based test environment.

In any scenario, one recommendation holds: **pair students as Player + Coach whenever feasible** (discussed more in 5.4). Even if you have enough devices for everyone, pairing up can be beneficial for collaboration and anxiety reduction. For instance, in a lab with 30 computers, you might still have students sit in pairs at 15 computers – one plays, one watches and then they swap. This can slow things down slightly, but the peer support sometimes leads to better outcomes for struggling students.

Another universal tip: **teach and enforce a consistent way for students to enter their Name, School, Class** when submitting . For class tracking to work, all students must enter the class name exactly the same way each time. It’s worth spending time initially to show “Type your Name as First Name and maybe last initial, Select School from dropdown, and for Class write ‘Std 4 Blue’ (for example)”. If the class naming is standardized, your class leaderboard will neatly show everyone’s times. If not, you’ll see split entries (e.g., “Std4 Blue”, “4 Blue”, “Standard 4 Blue” considered different classes by the system). Some teachers even assign a “class code” to ensure consistency. This small administrative detail pays off by making the data clean.

Finally, consider the **physical setup** of devices: If you have multiple devices, spread them out in the room to avoid students crowding or peeking at each other’s answers (though the random nature of multiplication grids means direct copying is unlikely, but still). If using a phone or single tablet, maybe designate a corner of the classroom as the “Fluency Station” where the student(s) go to play with minimal distraction.

By planning for your device scenario, you ensure that every student gets a fair chance to use CountMeInTT regularly. It's quite possible to start with a suboptimal scenario (like 1 device) and later move to a better one if your school acquires more devices or as other teachers see the success and coordinate resources. Indeed, seeing the engagement, some schools have invested in a couple more tablets for shared use. But the key takeaway is: **no matter how few or how many devices you have, there's a way to implement CountMeInTT effectively**.

5.4 Classroom Routines and Management (Pair System, Stations, etc.)

Introducing CountMeInTT to a class goes beyond just the tech – it's about setting up routines that maximize student engagement and minimize off-task behavior. Over time, a few **teacher-tested routines** have emerged that work in virtually every scenario. Two of the most effective are the **Player–Coach Pair System** and the use of structured **grid monitoring/feedback routines**.

The Pair System (Player + Coach): This simple routine involves pairing up students during the activity – one student is the "Player" (actually entering answers on the device) while the other is the "Coach" (sitting beside or behind, watching and supporting) . Here's why and how to implement it:

- **Why it works:** The Coach's role is to keep the Player calm, observe where the Player hesitates, and double-check that the Player enters their name/class correctly on submission . This accomplishes several things. First, it reduces anxiety – the Player knows a classmate has their back, and the activity feels more social and less like a solitary test. The Coach can whisper encouragement or help refocus if the Player freezes on a fact. Second, it keeps the waiting students (the Coaches) **actively engaged** rather than just idle. They have a job: watch and note any struggles. Third, it promotes collaborative learning – often the Coach picks up on patterns too, and the pair might discuss afterward ("I saw you slowed on the 7s, maybe practice those").

- **How to implement:** Pair a faster student with a slower student for balance, or simply do random pairings and rotate them periodically so everyone gets to be Player and Coach. Clearly explain the Coach's duties (not to give answers outright, but they can prompt or say "try 6×7 again" if the player is blanking, for example). Also, instruct the Coaches to be the ones to ensure the submission form is filled correctly – sometimes excited Players might type their name wrong or forget class, so a Coach double-checks before they hit submit. After submission, have them swap roles. If you're doing one attempt per week, you might actually keep roles consistent for that attempt and then next week switch roles. If doing two attempts in one session (a practice and official), you can have them swap for the second attempt.

- **Results:** Teachers who use the pair system often report a **calmer, more cooperative atmosphere** during the activity . Students adopt the language of coaching, saying things like "You got this!" or "Don't panic, skip that one and come back." This peer support is invaluable for students who might otherwise get flustered by the timer. Additionally,

coaches can help maintain discipline by gently correcting a peer ("No talking now, focus on the grid, I'll handle any question you have after"). It's training students to help each other – a nice side effect beyond math.

Grid Monitoring and Identifying Weak Spots: While students are playing, a teacher (and/or a Coach) can perform "grid monitoring," meaning they **observe where students slow down or struggle** to fill answers . This is a form of informal assessment:

- As you watch a student play, you might notice, for example, they pause a long time on 6×7 or they repeatedly mistype 8×8. Take a mental note or jot it down. Over a few weeks, you may see patterns such as "Many of my students consistently get stuck on 7s and 8s" – that tells you those facts need review in class.

- This diagnostic use of the game turns it into not only a fluency builder but also a **tool for targeting instruction** . Perhaps after a session, you spend 5 minutes doing a quick drill or game specifically on 7× and 8× tables because you observed many hesitations there. Or you might pull aside a student who struggled with the 12×12 section and give them a tip or two.

- One strategy is to have a **"reflection" after each attempt**: ask students "Which part was hardest? Which tables slowed you down today?" Coaches can chime in with what they observed of their partner. For example, a coach might say "I noticed she slowed on the 9×9 and 9×8." This encourages metacognition – students start to become aware of their own weak spots and then can focus practice on those.

- The platform itself doesn't explicitly tell you which facts were missed or slow (it just gives the final time), so this human element of observation is key for gleaning deeper insights. **Teachers have effectively used this to group students or differentiate**. If you find a subset of the class is weak on certain times tables, you can give them targeted practice or assign those specific tasks for homework. CountMeInTT thus serves as a light formative assessment tool each week.

Classroom Management during the activity: A few routines can ensure things run smoothly:

- Always do a quick **"tech check" and login** before timing starts. For example, open the site on all devices, ensure they are on the grid selection screen, and only then say "Go." This avoids some students fumbling with URLs while others started.

- Establish a norm that once a student finishes, they should **sit quietly or help their coach**. Perhaps they can quietly cheer others on or do a silent high-five, but to avoid chaos, encourage them not to shout out their times or disturb others still working.

- If doing rotations, use a **timer or signal** (like a bell) to indicate when to switch stations or pairs. Practice the transitions.

- Keep a visible **list or chart of who has done their attempt** (this can be a simple checklist on the board). Students love seeing their name checked off ("official attempt done for this week"). It also prevents the "Did I go already?" confusion.

- Introduce **mini-rewards or recognition** for good behavior during the routine. For example, you might say you'll award a point to the team/pair that shows the best focus or best sportsmanship (coaching). This can incentivize students to take the roles seriously.

Utilizing Team or Group Play: For variety, or when students seem to need a motivational boost, consider some **team-based routines**:

- **Relay Play:** Particularly with Standard 5 or combined classes, you can do a relay race where each student on a team completes a section of the grid and then tags the next person. The guide even suggests a relay across grid levels , e.g., one student does a 5×5, then the next does a 5×12, next 12×12, and the last does 15×15, and you add up their times or simply celebrate completion. This is more of a fun event than a weekly routine, but it can spice things up mid-term or be used as a "Fluency Championship" activity.

- **Group Competition:** If you have multiple classes or streams, you can pit class vs class (within the school or even across schools if using leaderboards). For example, announce that this month Std 4A and Std 4B will compete for best average time or most improved. Use the data from the platform or your trackers to declare a winner. This taps into inter-group competition which can be friendly and motivating (just ensure it stays positive).

- **Whole-class attempt projected:** Another routine for engagement is occasionally doing a whole-class attempt where one grid is projected and the **class works orally as a team** to fill it (with one student typing). This is not for official timing, but as a practice game: you could time how long the class takes collectively to fill a 12×12 on the projector by calling out answers. This promotes unity and allows discussion of strategies (like "We all know the 5s, fill those first!" or "Remember 9×9!"). It can also serve to demonstrate techniques like skipping a hard fact and coming back, etc.

Time-of-day integration: Decide *when* to integrate the routine. Many teachers find that doing it at the **start of a math lesson or day** is effective – it serves as a warm-up and gets the brain in gear. Others do it at the **end of math class** as a fun closing activity. Some even use it as a transitional activity after lunch or when students seem sluggish, as it is short and energizing. The best time might also depend on device sharing (if multiple classes share devices, you may have assigned slots). Whatever you choose, try to attach CountMeInTT to a consistent part of

the schedule (e.g., every Tuesday and Thursday after morning assembly, or daily right after lunch break) so it becomes a habitual routine.

Addressing off-task behavior or misuse: Since this is a game-like format, a few students might be tempted to play around (e.g., deliberately put wrong answers to see reds, or submit joke names). It's important to set expectations from the get-go that the game is still a **learning tool**: treat it with respect, use your real first name (or whatever format teacher says), and try your best. The vast majority follow this once they see the competitive angle (they want a good time, so they focus). If someone finishes extremely fast with suspicious accuracy (maybe they memorized a pattern or something), just have them do it again properly – but in reality, cheating the system is hard if they must get all correct. Some might try to refresh to avoid a bad time being submitted; you can discourage that by emphasizing improvement over time, not perfection in one go. And of course, remind them that **the goal is personal growth, not just being #1** – that helps reduce any urge to cut corners.

In conclusion, with thoughtful routines like the Pair System and active monitoring, you can run CountMeInTT sessions that are not only academically beneficial but also smooth and enjoyable from a classroom management perspective. Students stay engaged, help each other, and you as the teacher gain insight and maintain control. Over a few weeks, you'll likely find that the routine almost runs itself – students will know exactly what to do, and you might even hear them say, "Sir/Miss, can we do the table today?" as they eagerly anticipate their fluency practice session .

5.5 Integrating CountMeInTT into the Timetable

The final piece of the implementation puzzle is making sure CountMeInTT finds a **permanent home in your weekly timetable** and aligns with your school's academic schedule. Integration is key to sustainability – if it's haphazard ("maybe we'll do it if we have time"), it tends to fall off the radar when things get busy. Here are strategies to weave the fluency tool into your regular plans:

- **Assign a dedicated slot:** Treat the fluency session like any other class activity that has a fixed schedule. For example, if your timetable has a "Mental Math" period or a "Drill Practice" time, designate that as CountMeInTT time. If not, create a slot such as every Monday 10:00-10:10 or Fridays last 20 minutes. By writing it into your lesson plans or the timetable, it sends a message that this is a regular part of the curriculum, not an optional extra. Many teachers use the first 10 minutes of the first Math lesson of the week for this – it acts as a quick recap of basic skills to prime students for new content.

- **Coordinate with other teachers (for device sharing):** If you are in a school where devices (or computer lab time) are shared among classes, it helps to have a conversation in advance with colleagues or administrators to block a recurring time. For instance, you might arrange that Std 4 gets the tablets every Wednesday 11:00-11:30 for CountMeInTT, Std 5 gets them Thursdays, etc. This avoids conflicts and ensures you actually have the resources when you plan to. In one school pilot, teachers created a

simple rota that reserved the single laptop cart on different days for each class's fluency session – once that was done, it was easy to stick to the plan.

- **Use as part of Homework or Enrichment:** While our focus is classroom implementation, consider integrating CountMeInTT into your homework routine as well. You can encourage students (especially those with internet at home) to practice on their own and then report their best times in class. Some teachers have even made it a standing homework assignment: "Play CountMeInTT for 10 minutes and write down your best time of the week to bring to class." This can supplement the in-class routine and is especially useful if class device access is low; it offloads some practice to home (though be mindful that not all will have access, so keep it optional or bonus). If used as homework, remember to **recognize home effort** – e.g., "Student A practiced at home and improved 5 seconds, let's applaud that." This integration reinforces the idea that fluency is an ongoing goal, not limited to school hours.

- **Academic timetable alignment:** Tie the use of CountMeInTT to the curriculum topics you are teaching. For instance, when you're covering multiplication or division topics in class, highlight that the fluency practice complements those. During the geometry or measurement units, you might reduce frequency if needed, but emphasize that keeping up the fluency will help in calculations (like area, volume formulas requiring multiplication). CountMeInTT can also be aligned with the **SEA preparation period** – in the term leading up to SEA (usually Term 3 for Std 5), you might intensify usage (maybe two attempts per week or daily quick sessions) as part of exam prep. It's far more effective than just drilling past papers without addressing underlying fluency issues.

- **Integrating into School Culture:** Work with school leadership to integrate the program into broader school activities. For example, a principal might allow 5 minutes in assembly to mention which class had the highest participation or to give a shout-out to a "Student of the Week" for fluency. If there are bulletin boards, create a **Fluency Wall** where you update class records or top scores. This way it becomes part of the school's academic culture, similar to how some schools display reading counts or spelling bee winners. In Trinidad & Tobago, some schools have "Math Month" or similar events – that's a perfect time to showcase CountMeInTT (perhaps hold an inter-class competition or demonstration).

- **Plan around the school calendar:** Recognize that there will be times when normal routines are disrupted – e.g., exam weeks, sports day, holidays. Plan ahead for these. If end-of-term exams are coming and you can't spare class time, maybe that week skip the official attempt or have students do it at home. If school is closed on your usual day, perhaps do it earlier. The idea is to aim for consistency (like one attempt per week per student), but be flexible to not stress out over missing a session here or there. The key is to resume afterward and perhaps do a quick review to get back on track.

- **Bridging to other math practices:** Integrating also means linking the outcomes of CountMeInTT to your teaching. For instance, when teaching multi-digit multiplication, you can point out how knowing the single-digit facts quickly (thanks to CountMeInTT) makes the long multiplication steps easier. Or in fractions, converting 3/4 to a percentage is easier if you recall 3×25 = 75 quickly. By explicitly making these connections in class discussion, students see CountMeInTT not as an isolated game, but as a part of their math toolkit that is helping them across topics.

- **Encourage peer accountability:** Once integrated, students can help keep it going. If it's Wednesday and you somehow forgot to start the session, you might find students reminding you: "Miss, aren't we doing our times table challenge today?" In fact, this has been reported by teachers – students start to own the routine because they enjoy it and see their progress. This is the best kind of integration: when it becomes a class norm that even the kids protect. Cultivate this by occasionally asking a student to lead the routine ("Today, Alex will be our timekeeper and tell us when to switch" or "Maria will announce the top 3 improvements today"). That involvement cements it in their expectations.

In summary, integrating CountMeInTT into the timetable means giving it a rightful place in your teaching schedule, aligning it with curriculum and school events, and maintaining consistency. The payoff for doing so is huge: you get a class of students who steadily improve a critical math skill without you having to expend tons of teaching time or marking energy. It's a short investment each week that yields cumulative gains. The later sections will look at how to track those gains and assess the impact, but it's clear that only with regular integration can the tool achieve its full potential in improving math fluency and confidence.

6. Tracking Progress and Assessment

One of the advantages of CountMeInTT is that it naturally generates data on student performance. Every time a student completes a grid, you have a record of their time and accuracy (since only all-correct times count). This opens up possibilities for tracking progress, assessing student growth, and even assigning grades or awards in a fair way. In this section, we discuss how to use the platform's data and some provided tools (leaderboards, trackers, rubrics) to monitor progress and incorporate fluency achievement into your assessment plan.

6.1 Leaderboards and Data Tracking

Leaderboards as Motivation and Monitoring: The built-in leaderboards on CountMeInTT's website serve as a public (or at least school-wide) display of performance. For motivation, many teachers find that simply seeing one's name climb from, say, 50th to 20th on the national board, or from 5th in class to 2nd, is incredibly motivating for students. It taps into their competitive spirit in a healthy way. For teachers, leaderboards can be a quick monitoring tool: **at a glance, you can see the top times and who has improved**. However, leaderboards typically highlight

the fastest times, which means they might not show improvement of a slower student who is not yet near the top. Therefore, while leaderboards are great for recognizing high achievers and creating excitement, they should be complemented with other tracking to capture everyone's progress.

Class and School Leaderboards: If your students all input the school and class names consistently, you can utilize the **class-specific leaderboard** to track standings within your class. For instance, visiting countmeintt.com/leaderboard?school=YourSchool&class=Std4Blue (or a similar URL) might show just your Std 4 Blue class's results ranked by time. This class leaderboard can be printed or projected each week to discuss in class. Some teachers maintain a **printed leaderboard poster**: after each session, they update a chart on the wall with the three fastest times and the "most improved" (which they calculate from their records). A **printable leaderboard template** is provided in the teacher resources (essentially a blank table to fill in names and times) . Using this, you can track each week's best performers or simply list every student with their latest time. **Visually posting progress** makes it tangible; students can physically see their name moving up or their times decreasing, which reinforces their sense of accomplishment.

Progress Trackers for Individual Growth: In addition to leaderboards, a **Progress Tracker Sheet** for each student is highly recommended . This is usually a simple log where students (or the teacher) record each attempt's date and time. For example, a row might say: Week 1 – 5×12 grid – 4 min 50 sec; Week 2 – 4 min 20 sec; Week 3 – 3 min 55 sec; etc. By keeping such a log, a student can clearly see their journey. Many teachers print a one-page tracker for each student at the start of term, which the student keeps in their folder or math notebook, adding their new time after each attempt. This practice fosters **self-monitoring** – students take ownership of noting and reflecting on their performance . When a child sees their times dropping over weeks, it boosts confidence and makes the improvement concrete. If a time one week is slower, that can open a discussion ("What happened? Not feeling well? Okay, no problem, let's aim to bounce back next week."). The Progress Tracker Sheet is not for grading but for growth reflection.

Teachers can also keep a **class progress spreadsheet** if they prefer digital. The platform's data could be exported (the creator provides snapshots or you can manually enter times). In such a sheet, each row is a student and columns are week1 time, week2 time, etc. This makes it easy to calculate improvement or identify who hasn't improved. However, even manually, classes aren't that large, so updating times weekly on paper or a board is feasible.

Example of Tracking Data: Consider a Standard 4 class where at the start of Term 1, the range of times for 5×12 grid is between 10 minutes (slower students) and 3 minutes (faster students). By the end of Term 1, after weekly practice, perhaps the range is 5 minutes to 1.5 minutes. Every student has improved. A teacher's record might show Student A went from 10:00 to 6:40, Student B from 5:00 to 3:00, Student C from 3:30 to 2:50, etc. You might create a simple **bar chart of each student's first vs best times** to share with the class or at a parent meeting (this can be anonymized or not). Such visualization celebrates that growth is happening across the board.

In fact, during a pilot test phase (201–2025) with a small group of students, **every single student improved their speed, with an average improvement of around 50% faster completion time** . Some of the largest gains were over 70% improvement in time . For example, one student initially took over 20 minutes to complete a 12×12 grid and after consistent practice brought that down to about 6 minutes . Even students who started with very slow times (20 minutes) managed to finish under 6–7 minutes by the end of the practice period . Figure 1 below illustrates the percentage improvement of nine pilot students, all of whom showed substantial gains in fluency over several weeks.

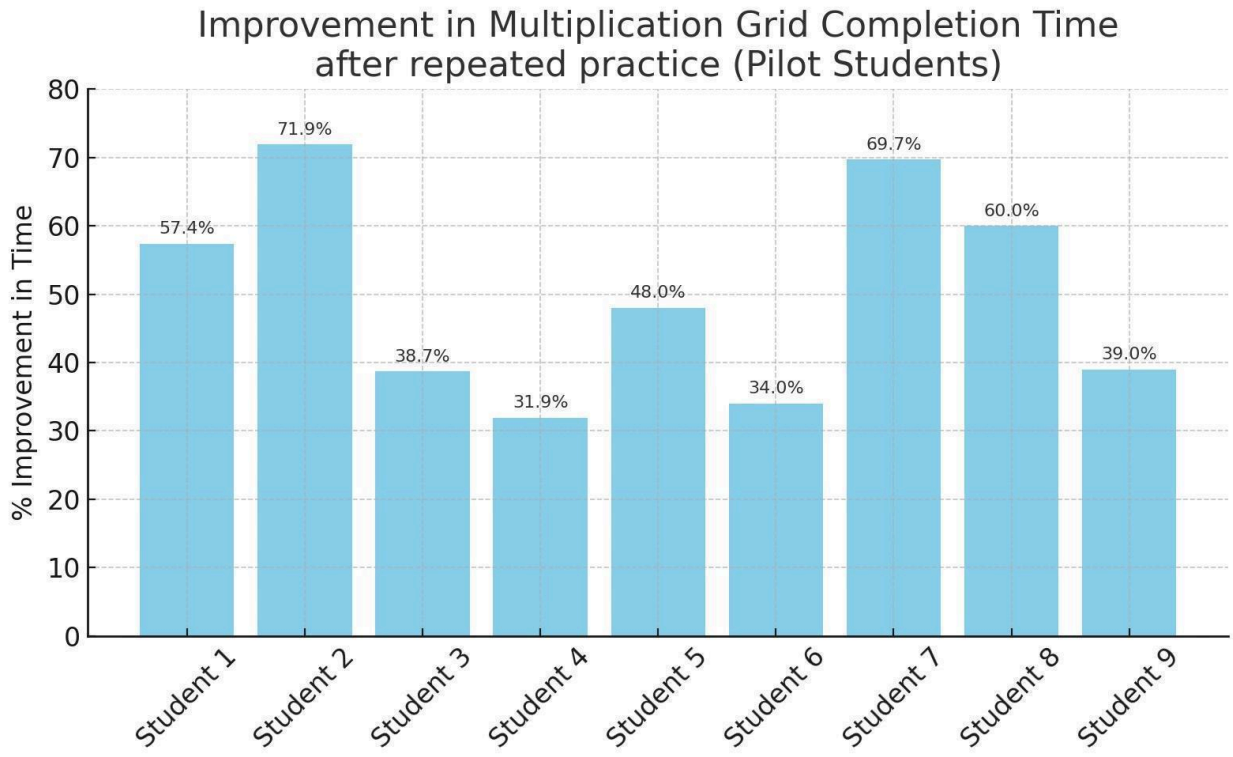

Figure 1: Percentage improvement in multiplication grid completion time for nine students after regular CountMeInTT practice. Each student saw significant fluency gains, with an average improvement of ~50% and some exceeding 70% .

This kind of data is powerful. It not only validates the effectiveness of the practice to skeptical stakeholders, but also can be shared with students: “Look class, on average you all are 50% faster than when we started – that’s amazing progress!” Students feel proud being part of that success story.

Data Snapshots and Reports: If you are part of a formal implementation (perhaps a pilot at your school), the CountMeInTT team might provide you with **monthly reports** summarizing usage and improvements for your class or school . For example, a monthly report might include total plays, distinct students, breakdown by class, and highlight top performances . If you receive such reports, use them to inform your teaching: maybe one class has fewer plays – perhaps they need more encouragement or a schedule adjustment; or you notice Standard 3

played a lot but improvement was modest – maybe intervene with some strategy coaching. If these snapshots are not automatically provided, you can always request data from the site admin (especially if students log with unique identifiers).

In essence, tracking progress with CountMeInTT can be as simple or as detailed as you like. At minimum, keep a record of each student's times – this will allow you to measure growth. Maximally, you can dive into analytics, compare class averages over time, or even correlate fluency gains with exam scores later. The data is there; harnessing it will make fluency practice feel more purposeful and will equip you with evidence of learning.

6.2 Growth-Focused Assessment (Rubrics and Recognition)

Assessing multiplication fluency doesn't have to mean giving a traditional test and marking it. With CountMeInTT, you have an opportunity to assess students in a **low-stress, growth-focused manner**. Many teachers choose to incorporate the fluency work into their grading scheme, not to emphasize speed as an "exam" but to reward **participation, improvement, and consistent effort**. Here we discuss a sample rubric and recognition strategies to assess fluency progress in a fair and motivating way.

10-Point Fluency Rubric: One effective approach is to allocate a small portion of students' math grade (or a weekly/term points system) to multiplication fluency via CountMeInTT. The guide suggests a **10-point rubric** that can be used either for weekly scoring or accumulated for term assessment :

- **4 points – Participation/Completion:** Did the student **complete** the grid with all answers correct (at least once in the period)? This basically rewards those who made a valid attempt and got through it, regardless of time. Earning all 4 points here might require, say, completing an official attempt each week of the month. You could also scale it (e.g., 3 points if they completed but needed assistance, 2 if they attempted but didn't finish, etc., but the simple way is full credit if completed).

- **3 points – Improvement vs. Personal Baseline:** Did the student **improve their time** compared to their own baseline (e.g., from the first attempt of term, or from last term)? This incentivizes students to beat themselves. If a student's time got better over the month/term, they earn these points. If a student was already very fast and maybe improvement is hard to see, you could award these if they maintained an already excellent performance or helped others – be flexible but document it. Typically, even fast students can improve a little or at least show consistency.

- **2 points – Consistency:** Did the student **show up every week** and give an effort (i.e. logged an attempt in X out of Y sessions)? For example, 2 points if they participated in all sessions, 1 point if they missed one or two, 0 if they were frequently absent or opted out. This emphasizes that doing it regularly is valued.

- **1 point – Effort/Conduct:** Did the student **stay on task and follow rules** during the fluency sessions? This is a behavior/effort metric. Almost all students will get this if they try without disrupting. It gives you a small lever to address off-task behavior ("you lost that 1 point this week because you were distracting others, let's improve that").

Using this rubric, each student can get up to 10 points for the fluency component . You might score this monthly or at the end of term. For example, you might announce that "Math Fluency" will count for 5% of the term grade, and it will be scored out of 10 using these criteria. The beauty of this rubric is that **even the weakest student, who may never be near the top of the leaderboard, can still earn a very good score by trying and improving** . It takes the pressure off raw speed and places it on growth and participation.

Fairness and Equity: To keep things equitable, it's advisable not to grade on absolute time (like "sub-3 minutes gets A, 6 minutes gets C" – that would penalize some kids unfairly). Instead, as described, grade on **completion and improvement**. This approach has been well-received because strong students still get rewarded (they'll complete and likely improve, so they get full points), and struggling students have a pathway to full points too (by just doing their best and getting a bit faster than before) . It reframes the activity from a race against others to a personal progress journey in the context of grading.

Some schools may prefer not to formally grade it but rather use it for **house points** or some internal rewards. For instance, completing your weekly attempt might earn your house or team a point in an ongoing competition. Or they might incorporate it into the effort grade or a math behaviors rubric.

Recognition and Rewards: In addition to (or instead of) formal grading, recognizing students' achievements in fluency can be very motivating. The guide and teacher feedback suggest recognizing two main categories: **Top Performers** and **Most Improved** . Each addresses a different group of students:

- **Top 3 (Fastest Times):** At the end of each week's session or month, congratulate the top 3 fastest students in the class (or in each class if sharing in assembly). This gives the speedsters their moment and can spur friendly rivalry for next time. You might give small stickers, certificates, or simply applause. It's crucial to handle this in a spirit of fun, not to imply that being fastest is the only thing that matters – which is why you pair it with…

- **Most Improved:** This is arguably even more important to celebrate. Identify which student(s) cut their time the most significantly compared to last week or last month. Often this will be a student in the middle or bottom of the pack who made a big leap, and highlighting them sends the message that improvement is valued as much as raw performance . For example, "Shanelle improved from 5 minutes to 3 minutes 30 seconds – amazing progress, she's our Most Improved this week!" The guide notes that "Most Improved is one of the best ways to reward weak-to-middle students without embarrassing them" . Indeed, a student who is never going to be in the Top 3 might, over

a month, quietly improve 40% – when you recognize that, their face lights up and others see that everyone has a chance to be praised.

- Other fun awards you can consider: **Most Consistent** (the student who always shows up and delivers steady times, maybe most weeks played as well) , **Personal Best Champion** (student who achieved a big personal best jump in one week) , **Fluency Star** (met a target time or a goal, like “broke the 2-minute barrier”) . These can be rotated or used occasionally to keep things interesting.

Rubrics in Practice: Let’s illustrate how a term assessment might look using the rubric:

Suppose by end of term:

- Student X participated every week, completed all attempts, initial time 8:00 -> latest time 5:00 (improved 3 minutes), always on task. They get 4 (completion) + 3 (improved significantly from baseline) + 2 (came every time) + 1 (good effort) **= 10/10**.

- Student Y participated every week, completed, but maybe started at 3:00 -> latest 2:45 (improved only a bit, but they were already fast). Still, that’s improvement (could give full 3 or maybe 2 if being strict that it wasn’t a large improvement). They were on task. So they get 4 + 2 + 2 + 1 = **9/10** (the message being, you did great, but you can’t slack – though if they truly had almost no room to improve, one could still give 3 to avoid punishing being good initially).

- Student Z missed one attempt (maybe absent one week), and improved slightly 10:00 -> 9:30, and needed some reminders to focus. They get perhaps 4 (they did complete at least one) + 2 (some improvement) + 1 (missed one week, so not full consistency) + 1 (effort acceptable) = **8/10**. This still reflects decent effort with note that more consistency or focus could help.

- Student W was often unfocused or had incomplete attempts (maybe needed help to finish or gave up once). They might get 2 (partial completion) + 1 (minimal improvement or hard to gauge) + 1 (inconsistent) + 0 (poor effort) **= 4/10**. This student would need intervention and encouragement, rather than just a bad grade – rubric scoring can flag them for support.

You see, the rubric helps differentiate among levels of engagement and progress, and it gives **actionable feedback** to students. Instead of just a score, you can say “I gave you 1 out of 2 on consistency because you missed two sessions; let’s try to attend all in the next term. And you got 2 out of 3 on improvement; maybe set a goal to improve by 20% next term to get full points there.”

Extra Credit via Fluency: Another approach if you don't want to include it in formal grading is to use fluency as a source of **bonus marks or enrichment credit** . For example, some teachers have a system where if a student improves their time by certain percentages, they get bonus points added to a test or some reward. The guide gives an example: +1 mark on a quiz if personal best improved by >20%, +2 marks if >35%, +3 marks if >50% over a period . This is entirely optional but it can be a safe way to incentivize practice without making it punitive. Students realize their fluency work can tangibly benefit them academically, yet if someone doesn't improve, it doesn't hurt their base grade, they just don't get the bonus.

Ensuring Grading Fairness: If you do include it in grades, a couple of policies keep it fair:

- Only count **in-class attempts** for grading . If students play at home, that's great for learning, but don't count a home result that others didn't have equal chance to do under supervision. Home practice can be encouraged for personal benefit, not for the grade.

- Keep the grade weight **modest** (5-10% of math grade) . It should complement, not overshadow, other assessments. This also alleviates parental concerns ("Are they just playing a game for marks?") – you can explain it's a small component aimed at skill-building.

- Emphasize **equity**: improvement and completion, not absolute speed . Communicate this policy to students and parents so they understand that everyone has the opportunity to score well if they put in effort.

Templates and Tools: The implementation guide provides a **ready-to-print rubric + tracker sheet** that you can use or adapt immediately . This likely includes a formatted rubric and a blank table for tracking attempts. Using such templates saves time and ensures consistency in how you evaluate students.

In summary, the assessment aspect of CountMeInTT should be approached as *assessment for learning* rather than a high-stakes evaluation. By using a rubric that values participation and growth, and by publicly recognizing improvement and effort, you maintain a **positive, supportive assessment environment**. This way, the faster students remain challenged (to maintain their rank or improve further) and the slower students remain encouraged (because their efforts are seen and can earn them accolades and points). The result is a classroom where fluency improvement is a team effort and a point of pride, not a source of shame or anxiety.

6.3 Supporting All Learners: Differentiation and Motivation

One of the best aspects of CountMeInTT is that it naturally meets each student at their level (since everyone is essentially competing against themselves). However, in any mixed-ability classroom, you'll find some students zooming ahead and others struggling to even complete the grid. How do we ensure the tool benefits **all learners**, including those who are weaker in math or easily discouraged? And how do we keep the strong students engaged and challenged? This

section addresses differentiation strategies and motivational techniques to include every student in the fluency journey.

For Students Who Struggle with Multiplication: Not every child will take to the timed challenge immediately. Some may find even a 5×5 grid daunting at first, or may have anxiety flare up. Here are ways to support them:

- **Start Small and Team Up:** If a student cannot yet complete a 5×12 grid on their own, let them start with a smaller target. It's perfectly fine to have some students do **5×5 grids** in a Std 3 or 4 class as a scaffold, while others do 5×12. The important part is improvement. You can gradually introduce them to bigger grids. Additionally, use **team play** as a bridge – for example, an anxious student can do the grid with a "helper" student or as part of a team of 2-3 where each takes some rows . The guide specifically suggests if students are anxious, start with team play (e.g., groups of 5 students solving one grid together with a helper to confirm answers) . This lowers pressure drastically while still giving practice. Over time, as confidence grows, move them to independent attempts.

- **Encourage, Don't Penalize:** Make it clear that it's okay not to be fast. Publicly and privately encourage those who are last to finish – "You finished it, that's what matters! You got them all correct, good job." It can help to time how long the class is willing to wait so slower students don't feel bad; for instance: "Class, we'll give everyone up to 10 minutes to finish, no rush." If a student truly cannot finish within a reasonable time, consider letting them stop at e.g. 10 minutes and record how many they got, then work from there next time. But usually, they will finish if allowed to correct errors (since it's not a test with a strict cutoff, they can take even 15 minutes if needed on their first try).

- **Identify Weak Spots and Remediate:** Use the observation strategy (grid monitoring) to find which facts trip up the weaker students . Often, struggling students have specific gaps (like they never memorized 6×7, 8×9, etc.). Provide extra help on those – maybe give them a multiplication table chart to study, or practice those few facts with flashcards outside of the game. One can even use class time for a quick group drill on a fact family that many find hard (like the 7s).

- **Utilize "Helper" Peers:** Pair struggling students with patient, kind "coach" partners (not drill sergeants!). A good coach can reassure and give hints in a friendly way. For example, if the player is stuck, a coach might say "Remember 6×8, we did that one yesterday, think of 5×8 + 8…" – a gentle prompt. This peer tutoring element can make a world of difference. Just be sure to train coaches to prompt, not directly answer everything, so the player still does the retrieval.

For High-Flying Students: Some students will quickly dominate the grid times. They might consistently finish a 5×12 grid in under 2 minutes, or a 12×12 in 3 minutes, etc. To keep them engaged:

- **Introduce Harder Grids or Personal Challenges:** If a student has mastered 5×12, have them try 12×12 regularly. If they master 12×12 (e.g., under 4 minutes), introduce the **15×15 Pro grid** for a challenge . This larger grid includes facts beyond the standard curriculum (13×, 14×, 15×) which essentially tests how well they can extend their knowledge or do quick additions (since 13× is 10× + 3× etc.). It adds endurance because it's 225 cells instead of 144. Many top students love this "elite" challenge and it keeps them from getting bored. You can make a little deal: "If you break X time on 12×12, you qualify for the 15×15 challenge next time."

- **Emphasize Role as Leaders:** Strong students can be given leadership roles, like being **chief coach** or helping to track class data. This gives them a sense of responsibility beyond just being fastest. They could help update the leaderboard poster or tutor a weaker peer in learning a trick for the 9s table. By doing so, they remain invested even if the regular grid is no longer difficult for them.

- **Time Targets and Speed Control:** Very fast students can work on **speed control** – that is, maintaining accuracy at high speed. Sometimes in their rush to be ultra-fast, they make careless mistakes (which the system forces them to fix, slowing them down). Challenge them to get a "perfect run" with no mistakes, or to beat their own record by smaller and smaller margins. Introduce the concept of **"accuracy first, then speed"** if needed. This ensures they don't develop sloppy habits just to shave seconds. It's like training a runner to pace then sprint.

- **Leaderboard Recognition:** If your class's top student is also among the top nationally, celebrate that! It's a big deal to be e.g. the fastest Std 5 in the country on a grid. That kind of recognition can motivate them to keep polishing. Perhaps you can have a special **Hall of Fame** mention for any student who hits a certain milestone (like under 1 minute on 5×5, or top 10 national ranking, etc.).

Maintaining Motivation for All: Here are additional tips to keep the whole class motivated over time:

- **Set Class Goals:** In addition to individual goals, set a collective goal. For example: "If the class average time falls below 5 minutes by end of term, we'll have a pizza party," or "Let's see if we can accumulate 500 minutes total practice this month." Whole-class targets can foster team spirit – the faster help the slower to achieve it. Make sure goals are realistic and inclusive (not "all get under 2 min" which is impossible for some).

- **Variety in Routine:** While consistency is important, small variations can rekindle interest. Perhaps one week you do the session in a different venue (like outside with laptops, or in the library) just for a change. Or occasionally play soft instrumental music in the background while they compete (some students like this, others prefer silence – gauge your class). Or once, do a "backwards day" where coaches play and players

coach, just for fun.

- **Connect to Real Life:** Remind students why fluency matters. Have them notice in regular math class: “See how quickly John did that long multiplication because he knew 7×8 instantly? That’s what this practice helps with.” Or relate it to secondary school: “In Form 1, your teachers won’t let you use calculators for basics – you’ll shine if you already have this down.” Some students are motivated by knowing the practical benefit.

- **Parental Involvement:** Share progress with parents in a positive way. For example, send a note or mention at PTA that “Our class has collectively improved their multiplication fluency by 40% this term!” Encourage parents to ask their kids about their times or even to try the game themselves at home. If a parent tries it and sees it’s not trivial to get a fast time, they may appreciate their child’s effort more. Also, sometimes a little at-home competition (kid vs parent) can motivate the child to practice more.

- **Addressing Anxiety:** For the few who have deep math anxiety or performance anxiety, continue to emphasize **growth over ranking**. If needed, allow them to do an attempt privately (maybe during recess or with you alone) until they feel okay doing it with others. Ensure that in class nobody is mocking anyone’s times – cultivate a supportive atmosphere where everyone claps when the slowest student finally finishes. When students see genuine support from peers, it reduces their fear.

- **Differentiation by Role:** Not all students must engage with CountMeInTT in the exact same way every time. For instance, a student who is temporarily struggling in math could take on a timekeeping or data-recording role for a session instead of playing, just as a breather, then rejoin. Or a hyper-competitive student who’s getting frustrated by plateauing might be tasked to mentor a junior student which refreshes their perspective.

After the Fluency Routine: Use some reflection or discussion to consolidate motivation. Ask questions like “Who felt more confident multiplying numbers in other work this week because of this?” or “What strategy did you use to improve your time?” Get students to share tips (like “I fill all the easy ones like ×1 and ×10 first, then do the rest”) – this creates a collaborative learning environment rather than pure competition. It also surfaces learning strategies that others can try.

Inclusion is key: **every student should feel that CountMeInTT is helping them get better and that they are part of the class’s success story**. When implemented with sensitivity, the weakest students gain confidence and the strongest gain humility and leadership, and everyone gains skills. As evidence from local trials suggests, even students who initially showed **frustration or disinterest in math began to exhibit enthusiasm** and would ask to play more . They started viewing the multiplication practice as a friendly challenge rather than a dreaded test, often saying things like “Sir, can we do the table today?” excitedly . That transformation in attitude – from math anxiety to eager participation – is perhaps the most valuable outcome of all, and it arises only when we support and motivate every learner in the journey.

7. Impact and Evidence of Effectiveness

CountMeInTT is rooted in the belief (and now growing evidence) that consistent, focused practice can produce meaningful improvements in students' math fluency and confidence. After implementing the program in your classroom, it's natural to ask: *What is the impact? How do we know this is working?* In this section, we summarize findings from initial implementations and pilots, and how you can gauge the impact in your own context. We blend quantitative results (like improved times) with qualitative observations (like changes in student attitude), both of which are important indicators of success.

Fluency Gains: The most direct measure of impact is the improvement in completion times for multiplication grids. As discussed in section 6.1, pilots from 2021–2025 in Trinidad & Tobago have shown **100% of participating students improving their times**, often dramatically . In one tracked group of nine students over several months:

- Every student's final attempt was faster than their first attempt – some shaved off only a few minutes, others more than 10 minutes.
- The **average improvement was roughly 50%**, meaning on average students ended up completing the grid in half the time they initially took .
- Two students achieved around **70% improvement** – for instance, one went from about 20 minutes down to about 6 minutes (a 14-minute reduction) .
- Even the student with the smallest improvement still improved by ~30% .

This level of improvement suggests that regular practice with feedback truly does build fluency. It's worth noting that these were not necessarily top students to begin with; some started quite slow (taking 12-20 minutes) and yet were able to bring their times into a reasonable range (5-7 minutes) after repeated attempts . Essentially, **students who initially struggled to recall facts eventually were able to complete a full times table grid in a few minutes**, which is a significant achievement in terms of automaticity.

Your own class might show varying degrees of improvement, but you should expect to see a clear trend of decreasing times over weeks. One way to illustrate impact is to chart the class average time per week. Often, it yields a downward curve: maybe starting at, say, 8 minutes average and coming down to 4 minutes average by term's end. Another metric is **completion rate** – initially some students might not finish in the allotted time, later nearly all do finish within, say, 5 minutes cap. Additionally, look at **accuracy** – are students making fewer errors (thus fewer red corrections needed) as weeks go by? Many teachers report that not only do times drop, but the fluency practice carries over to fewer calculation mistakes in regular classwork.

Attitude and Confidence Shifts: The numbers only tell part of the story. Teachers have observed profound changes in how students approach math after using CountMeInTT regularly. Some anecdotal but powerful evidence from the pilot:

- Students began to show **excitement and enthusiasm** for something they used to dread. They would ask to play, even during free time . What started as a practice chore became "a friendly challenge they looked forward to" .

- There was a notable **boost in confidence**. Students who were shy about answering in math class started to participate more, possibly because their success in the game made them feel more competent. One student might think, "If I can handle the multiplication grid under time, I can handle these math questions too." Teachers reported kids using words like "fun" to describe times tables – a word rarely associated with drills before.

- **Resilience and perseverance** increased. At first, some students might have given up when stuck. Over time, they developed a mindset of "I can get it if I keep trying," since the game structure teaches them not to quit until all answers are correct. One teacher described that students grew more patient and focused in problem-solving in general, not panicking as quickly when faced with a tough question.

- Students started to develop a **healthy competitive camaraderie**. For example, they'd congratulate peers on personal bests, race each other during break for fun, or form little rivalries that pushed them to practice more. This created a communal sense that "we are all getting better together," especially when class leaderboards are used in a team spirit.

- Interestingly, some students began to show leadership as a result of the program's implementation. The best performers often helped the slower ones, creating a **peer-support culture** around math practice. Math fluency became "cool" in a way, because it was gamified.

In essence, the platform helped transform the narrative around math facts from "boring or scary drill" to "exciting challenge and chance to succeed." Given how crucial attitude is in learning, this is a significant impact. Research often notes that *math anxiety* and negative attitudes can inhibit student achievement; by alleviating these through positive experiences, CountMeInTT indirectly sets the stage for better outcomes in all math areas.

Performance in Formal Assessments: While there isn't yet large-scale data linking CountMeInTT usage to exam scores, it stands to reason that improved fluency will help in tests like SEA which are time-bound and require quick calculation. At the very least, students report feeling more ready and less stressed in timed exam sections that involve arithmetic. You might notice in class tests that students who used to leave multiplication-heavy problems blank now attempt them because they're less intimidated by the calculations.

If you track something like a pre-test and post-test on basic facts (e.g., a one-minute paper multiplication quiz), you'll likely find tangible improvements. Many teachers do a **baseline timed test on paper** at the start (say, "how many of these 50 multiplication facts can you do in 2 minutes") and then again later – the scores usually rise significantly after regular CountMeInTT practice. This can be useful data to present to school leaders or parents to show skill growth in a conventional format.

School-wide and Community Impact: In schools where multiple classes adopt the program, there's potential for a broader impact:

- **Math culture in the school improves.** Students talk about beating times and leaderboards the way they might talk about sports scores. It becomes a positive fad to know your times tables. For example, a principal from one pilot school noted that even during breaks, kids would borrow the computer lab to challenge each other – essentially choosing to practice math for fun.

- **Recognition in assemblies or newsletters** can highlight math achievements just like sports or arts. "This month our school's top CountMeInTT performers are…" – it adds prestige to academic effort.

- If neighboring schools or district officials see the success, it could lead to **broader adoption or competitions**. Trinidad & Tobago could even host inter-school multiplication bee-like events leveraging the tool, which further elevates the focus on foundational numeracy in the educational discourse.

- Importantly, an improvement in fluency across many classes could reflect in **national exam improvements** over time, especially in the basic operations sections. While a single class's data is small, collectively, if many schools use it, one might expect a gradual rise in primary math proficiency indicators.

Teacher Workload and Efficacy: From the teacher's perspective, an impact to note is the reduction in marking and planning for this aspect of math. Many teachers find they spend *less time* drilling multiplication in class or preparing extra worksheets, because CountMeInTT takes over that role efficiently. The weekly snapshots or your own tracking give you the information without grading piles of timed quizzes. This frees up teacher energy to focus on teaching higher-order skills and addressing conceptual issues, knowing that the routine is handling the rote practice in the background. This contributes to teacher morale and a sense of efficacy – you feel you have a handle on students' basic skills progress without burning out.

Anecdote: One student wholacked confidence in math, over the term, they diligently improved from 15 minutes to 7 minutes on the grid. At the end of term, during a session, this student proudly said, "Sir, I think I don't hate math anymore." For that tutor, that was the true victory – a change in mindset. The student went on to perform adequately in SEA (not top of class, but solidly), which the parent attributed to the boost in confidence and recall.

In conclusion, the impact of implementing CountMeInTT can be seen in **quantitative improvements (speed and accuracy)** and **qualitative enhancements (confidence, enthusiasm, reduced anxiety)**. As an educator, you should monitor both. Keep the evidence – perhaps maintain a portfolio with charts of class progress, samples of student tracker sheets, and notes on student comments. This not only helps in reporting to stakeholders but also in refining your practice. If something isn't working (say, a group of students plateaus), you can adjust strategy (maybe increase frequency or add incentives).

The results in Trinidad & Tobago so far are very promising: improved basic skills and a revitalized attitude toward math. By following the guide and tailoring it to your classroom, you are likely to witness similar positive outcomes. It reinforces the idea that with the right approach, **the foundational "times tables problem" in our schools is a solvable one** – we can build a generation of students who see math as a challenge to conquer, not a burden to fear.

8. Conclusion and Next Steps

Mathematics fluency, particularly in multiplication, is a cornerstone of primary education that opens the door to higher-level math success. The *CountMeInTT Comprehensive Teacher Implementation Guide* has presented a detailed framework for addressing this fluency challenge in Trinidad and Tobago's classrooms. By blending research-backed strategies with practical classroom routines, CountMeInTT offers a path to transform the way students master their times tables – turning a once tedious or anxiety-inducing exercise into an engaging, confidence-boosting game.

Let's recap the journey we've outlined: We began by understanding the **local problem** – many students were struggling with quick recall of multiplication facts, affecting exam performance and causing math anxiety. We surveyed **research findings** that emphasize the importance of fluency and the need for positive math experiences, which set the stage for why a tool like CountMeInTT is valuable. We then delved into what CountMeInTT is: its **philosophy and design** centered on timed grid challenges, personal bests, and leaderboards, all engineered to harness principles like retrieval practice, spaced repetition, and gamification.

The guide provided concrete **implementation strategies**, from selecting appropriate grid levels for each Standard to managing various device scenarios. Whether a teacher has one shared tablet or a full computer lab, there's a viable routine – be it a daily 10-minute rotation or a weekly 30-minute class challenge. We emphasized supportive **classroom routines** like the player-coach pairing and observation techniques that turn the game into a diagnostic tool for weak spots. We also integrated this routine into the broader **academic schedule**, suggesting ways to make it a seamless part of weekly planning and even homework.

Crucially, we addressed **assessment and tracking**. By using growth-oriented rubrics and celebrating improvement, teachers can grade or reward students in a way that motivates everyone. The data generated by CountMeInTT – whether in the form of times, leaderboards, or improvement percentages – becomes a powerful feedback loop to inform instruction and

recognize progress. We highlighted how to support **all learners**, differentiating for those who struggle and those who excel, ensuring that no student is left behind or held back by the program's design. CountMeInTT is not a one-size-fits-all blunt instrument; it's a flexible tool that, when wielded thoughtfully, can uplift each student at their own pace.

The **evidence of effectiveness** we discussed is heartening. From pilots and initial implementations, students are not only getting faster at multiplication, but also developing a more positive attitude towards math. They are proving to themselves that they can improve through practice – a lesson that extends beyond math facts to any skill in life. Teachers, meanwhile, are finding that this routine eases the burden of drilling basics and frees time to focus on deeper conceptual teaching, knowing that the fluency piece is being handled in a fun, student-driven way.

So, what are the next steps after reading this guide? Here are a few to consider:

- **Get Started:** If you haven't already, try CountMeInTT yourself. Visit the website, play a grid, and familiarize yourself with the interface. Then introduce it to your class with a sense of excitement. The first session can be exploratory – let them play around and get a feel for it.

- **Set up Your Routine:** Decide on which routine model fits your class and schedule. Mark it on your timetable. Prepare any materials needed (tracker sheets, a spreadsheet, sign-up for the leaderboard pages, etc.). If needed, talk to your IT coordinator or whoever manages devices to ensure internet access and device readiness.

- **Communicate with Stakeholders:** Explain the program to your students – why you are doing it and how it will benefit them. Also, inform parents at a PTA or via a note about this new initiative in class. When parents hear "free web-based math tool" and "practice for SEA", they're likely to be supportive. Share the rubric or how you'll acknowledge progress, so they know it's serious but also student-friendly. If you're a math lead or HOD, consider presenting this plan to your principal or colleagues, possibly encouraging a grade level or entire school to adopt it for consistency.

- **Monitor and Iterate:** Once you implement, watch how it goes for a couple of cycles. Are the kids engaged? Is the schedule working? Perhaps you find you need 15 minutes instead of 10, or pairing needs adjustment. That's okay – refine the process. The guide's advice is robust, but every classroom has its nuances. Use the data and your observations to tweak. For example, if one or two students aren't improving at all, intervene early (maybe they need a refresher of basic strategies or more encouragement).

- **Leverage the Community:** Connect with the CountMeInTT community if possible. There may be an online forum, or even local teacher groups using it, given it's a TT-based program. Sharing experiences can yield new ideas (like a teacher might share

an innovative reward system they used, or a new way to do teams).

- **Expand and Innovate:** After you've run it for a term, consider expanding its use. Maybe involve another subject teacher (some schools used it in house competitions), or tie it into a math club activity. You could even conduct a little **action research** of your own – for example, compare your class's fluency pre/post or compare to a class not using the tool if that exists, and document the outcomes. This could contribute to a larger evidence base and you might present it at a teacher conference or in the school newsletter.

- **Celebrate Success:** At the end of the year (or after SEA for Std 5), celebrate the fluency achievements. Maybe give out certificates to all students showing their personal best and improvement. This not only rewards them but also archives the progress made. It's a nice piece for their student portfolio and something they can be proud of.

In closing, **CountMeInTT is more than just a game – it's a strategic tool** to build a stronger foundation in math for our students. It aligns well with the curriculum goals of fostering computational fluency and it does so in a way that resonates with today's learners who enjoy interactive and competitive elements. By implementing this program, you are not only likely to see test scores inch up, but you will also be nurturing more confident, engaged, and resilient math learners. Students who no longer freeze at the sight of a multiplication problem, but instead tackle it with assuredness gained from practice.

The guide has equipped you with a comprehensive plan to implement CountMeInTT effectively. Remove the elements that don't serve your classroom, amplify those that do, and always keep the focus on **why** we do this: so that every child can say, "I know my times tables and I'm not afraid of math." With that mindset, they will be better prepared for the challenges of secondary education and beyond.

We encourage you to take the leap and make fluency practice a fun and integral part of your teaching. As you do, remember the core message: *every student can improve, every student can achieve fluency*. CountMeInTT is simply a means to that end – a means that has proven to work and to delight. Here's to seeing your students' fluency soar and their love for math grow. Happy counting, and count us in for your success!

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- **CountMeInTT Pilot Data (2021–2025)** – Internal reports by A. Burton. (Unpublished data from pilot implementations in Trinidad primary schools, indicating average 50% improvement in multiplication grid completion times and positive shifts in student attitudes)

Sample Leaderboard Screenshot